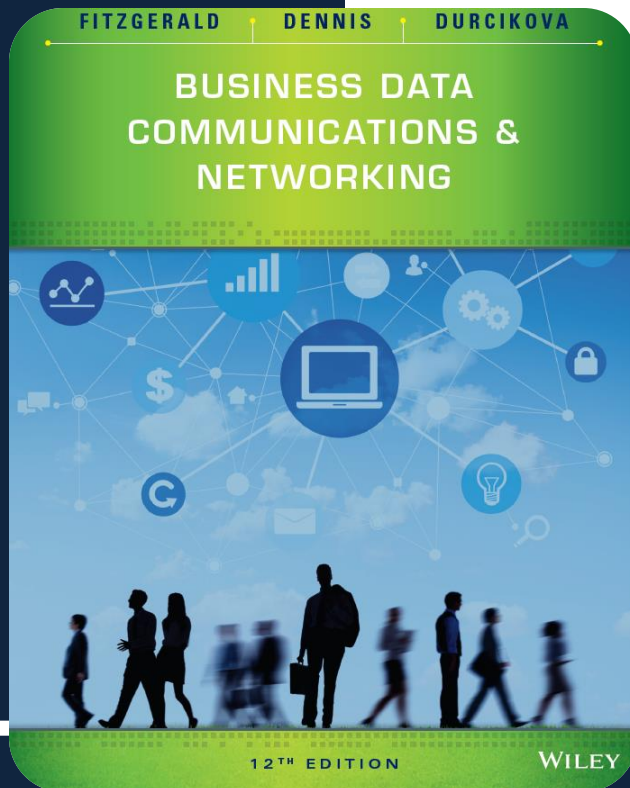




BUSINESS DATA COMMUNICATIONS & NETWORKING

Chapter 5 Network and Transport Layers

FitzGerald • Dennis • Durcikova

Outline

- Transport Layer Protocols
- Network Layer Protocols
- Transport Layer Functions
 - Linking to the application layer
 - Segmenting
 - Session Management
- Network Layer Functions
 - Addressing
 - Routing
- TCP/IP Examples
- Implications for Management

Network and Transport Layers

- Transport Layer
 - Layer 4 in the Internet model
 - Links application and network layers
 - Responsible for **segmentation** and **reassembly**
 - Session management (end-to-end connection between sender-receiver)
- Network Layer
 - Layer 3 in the Internet model
 - Responsible for **addressing** and **routing** of messages.

Internet Model

Application

Transport

Network

Data Link

Physical

Network and Transport Layers

- Transport Layer

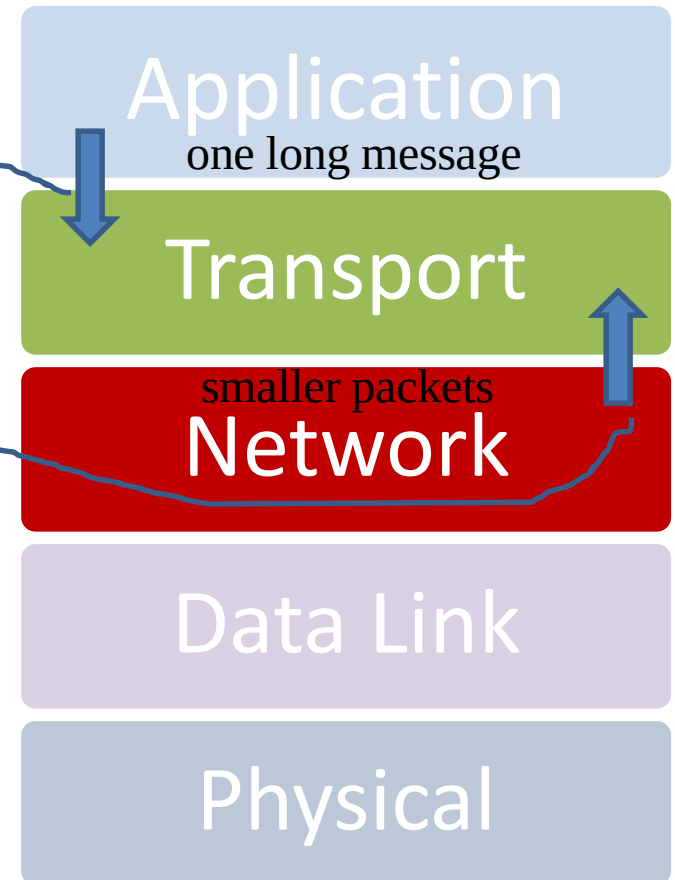
Segmentation (Packetizing)

Take outgoing message from the application layer and break it into a set of smaller packets for transmission through the network.

Reassembly

Take the incoming set of smaller packets from the network layer and reassemble them into one message for the application layer.

Internet Model



Network and Transport Layers

- Network Layer

Routing

Process of determining the path through the network that a message will travel from the sending computer to the receiving computer

Addressing

Finding address of next computer

Internet Model

Application

Transport

Network

Data Link

Physical

Protocols

- TCP/IP (Transmission Control Protocol / Internet Protocol)
 - Originally developed as a single internetworking protocol by Vint Cerf and Bob Kahn in 1974
 - Later divided into the TCP and IP protocols
 - Most common protocols of the Internet and in LANs, WANs, and backbone networks

TCP performs packetizing: Breaking the data into smaller packets, numbering them, ensuring each packet is reliably delivered, and putting them in the **proper order** at the destination. (Transport layer)

IP performs routing and addressing. IP software is used at each of the intervening computers through which the message passes; it is IP that routes the message to the final destination. (Network layer)

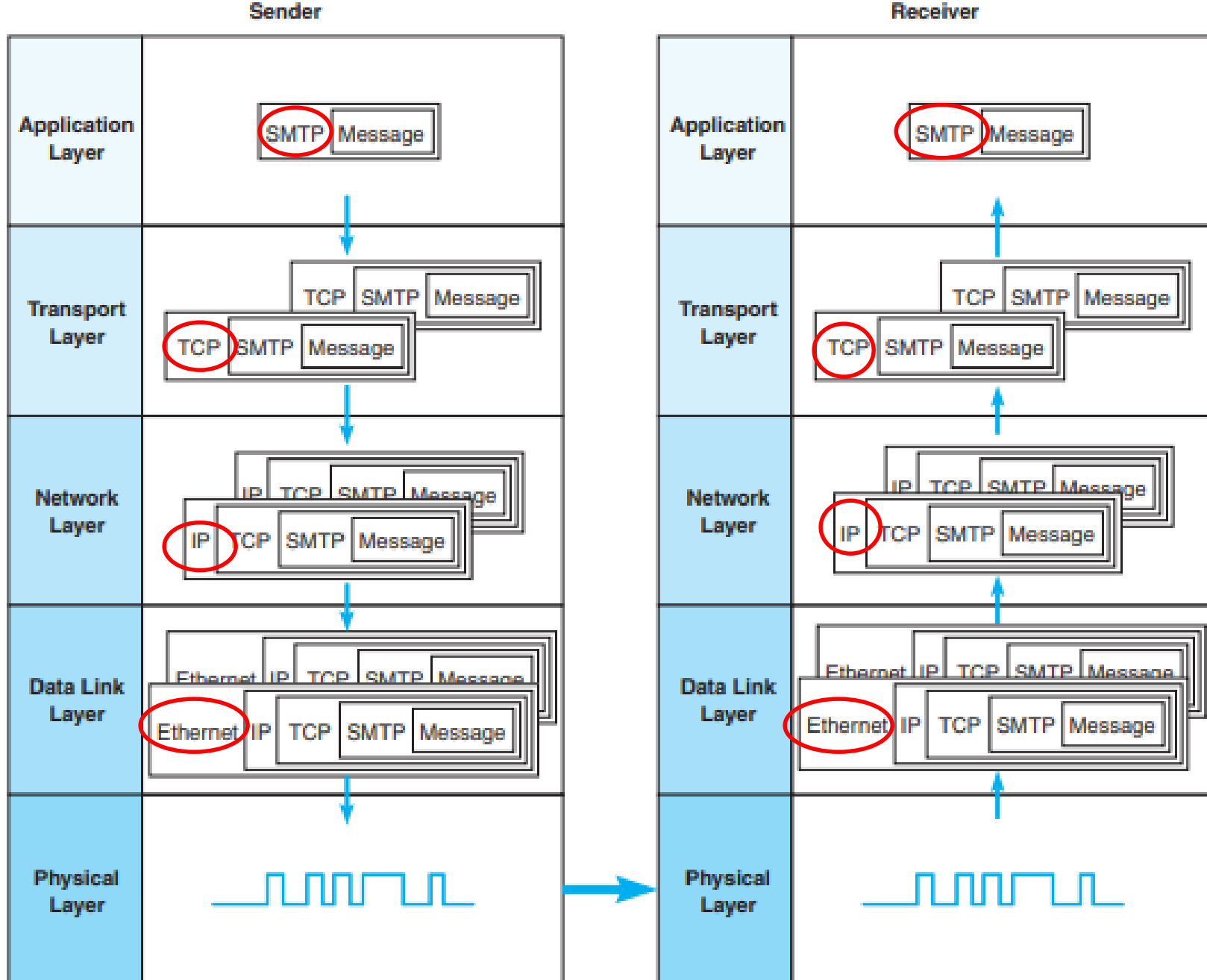


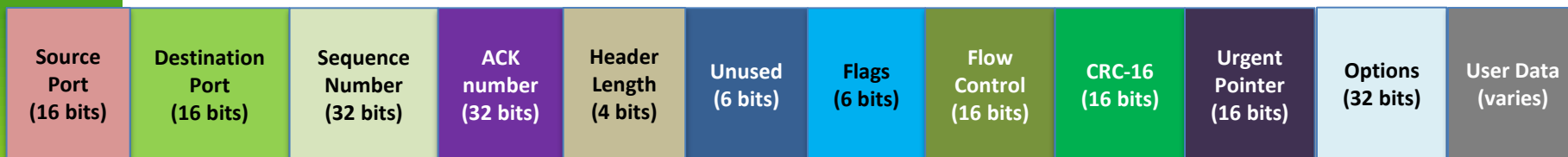
FIGURE 5.1 Message transmission using layers. HTTP = Hypertext Transfer Protocol; IP = Internet Protocol; TCP = Transmission Control Protocol

Transport Layer Protocols

- Transmission Control Protocol (TCP)
 - Connection-oriented protocol
 - Most common transport layer protocol
 - Used for reliable transmission of data
 - 160 - 192 bits (20 -24 bytes) of overhead

TCP performs packetizing: Breaking the data into smaller packets, numbering them, ensuring each packet is reliably delivered, and putting them in the proper order at the destination. (Transport layer)

TCP software **only** needs to be active **at the sender and the receiver**, because TCP is only involved when data comes from or goes to the application layer.



Transport Layer Protocols

Connection-oriented routing

- Sets up a temporary virtual circuit between sender and receiver.
- Network layer makes **one routing** decision when the connection is established, and **all packets follow the same route**.
- All packets in the same message arrive at the destination in the **same order** in which they were sent.
- **Packets only contain information about the stream** to which it belongs.
- Although many connection-oriented protocols include a sequence number to ensure that all packets are actually received, **sequence numbers are not needed**
- Connection-oriented routing **has greater overhead** than connectionless routing, because the sender must **first “open” the circuit** by sending a control packet that instructs all the intervening devices to establish the circuit routing. Likewise, when the transmission is complete, the **sender must “close” the circuit**.

Transport Layer Protocols

- User Datagram Protocol (UDP)
 - Connection-less protocol
 - Operates at the transport layer
 - Used in **time-sensitive** situations, for control messages, or when reliability is handled by the application layer
 - 32-64 bits (4-8 bytes) of overhead
 - Source port is optional in IPv4 and IPv6
 - Checksum is optional in IPv4



Transport Layer Protocols

Connectionless routing

- Each packet is treated separately and makes its own way through the network.
- Different packets may take different routes through the network depending upon the type of routing used and the amount of traffic.
- Packets following different routes may travel at different speeds, they may arrive out of sequence at their destination.
- Sender's network layer puts a sequence number on each packet, in addition to inform about the message stream to which the packet belongs.
- Receiver's network layer must reassemble them in the correct order before passing the message to the application layer.

Network Layer Protocols

- Internet Protocol (IP)
 - IP version 4 (IPv4)
 - Most common version of IP used
 - 32-bit addresses (2^{32} or ~4.29 billion possible)
 - Exhaustion of address space
 - IP version 6 (IPv6)
 - 128-bit addresses (2^{128} or $\sim 3.4 \times 10^{38}$ possible)
 - Slowly being adopted due to IPv4 exhaustion

IP performs routing and addressing. IP software is used at each of the intervening computers through which the message passes; it is IP that routes the message to the final destination

Network Protocols

- Each organization must assign the IP addresses it has received to specific computers on its networks.
- IP addresses are assigned so that all computers on the same local area network have a similar addresses.

Example

University : **128.184**.x.x (IPv4 , 32-bit addresses)

Business School LAN : 128.184.**56**.x

Computer Science LAN : 128.184.**55**.x

- Likewise, all the other LANs at the university that connects them, would have a different set of numbers.
- LANs are called **TCP/IP subnet** because they are logically grouped by IP number.
- Knowing if a computer is on your subnet or not it important for message routing.

Network Protocols

- IPv4 Packet

160-192 bits (20-24 bytes) of overhead

Options field rarely used

Optional Headers

Destination options

Routing

Fragment

Authentication

Encapsulation Security Payload

Mobility

Version number (4 bits)	Header length (4 bits)	Type of service (8 bits)	Total length (16 bits)	IDs (16 bits)	Flags (3 bits)	Packet Offset (13 bits)	Time to Live / Hop Limit (8 bits)	Protocol (8 bits)	CRC-16 (16 bits)	Source Address (32 bits)	Destination Address (32 bits)	Options (32 bits)	User Data (varies)
----------------------------	---------------------------	-----------------------------	---------------------------	------------------	-------------------	----------------------------	--------------------------------------	----------------------	---------------------	-----------------------------	----------------------------------	----------------------	-----------------------

- IPv6 Packet

320 bits (40 bytes) of overhead

Version number (4 bits)	Traffic Class / Priority (8 bits)	Flow Label (20 bits)	Payload length (16 bits)	Next Header (8 bits)	Hop Limit (8 bits)	Source Address (128 bits)	Destination Address (128 bits)	User Data (varies)
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Transport Layer Functions

1. Linking to the application layer
 - TCP/UDP may serve **multiple application layer protocols**
 - **Ports** used to **identify application** (2-byte numbers)
 - Many source/destination ports follow standards
 - Common port standards
 - HTTP: TCP port 80
 - HTTPS: TCP port 443
 - FTP: TCP ports 20 and 21
 - SMTP: TCP port 25
 - IMAP: TCP port 143
 - POP3: TCP port 110 (more commonly TCP port 995 secure version)
 - DNS: TCP or UDP port 53 (most commonly UDP)

Transport Layer Functions

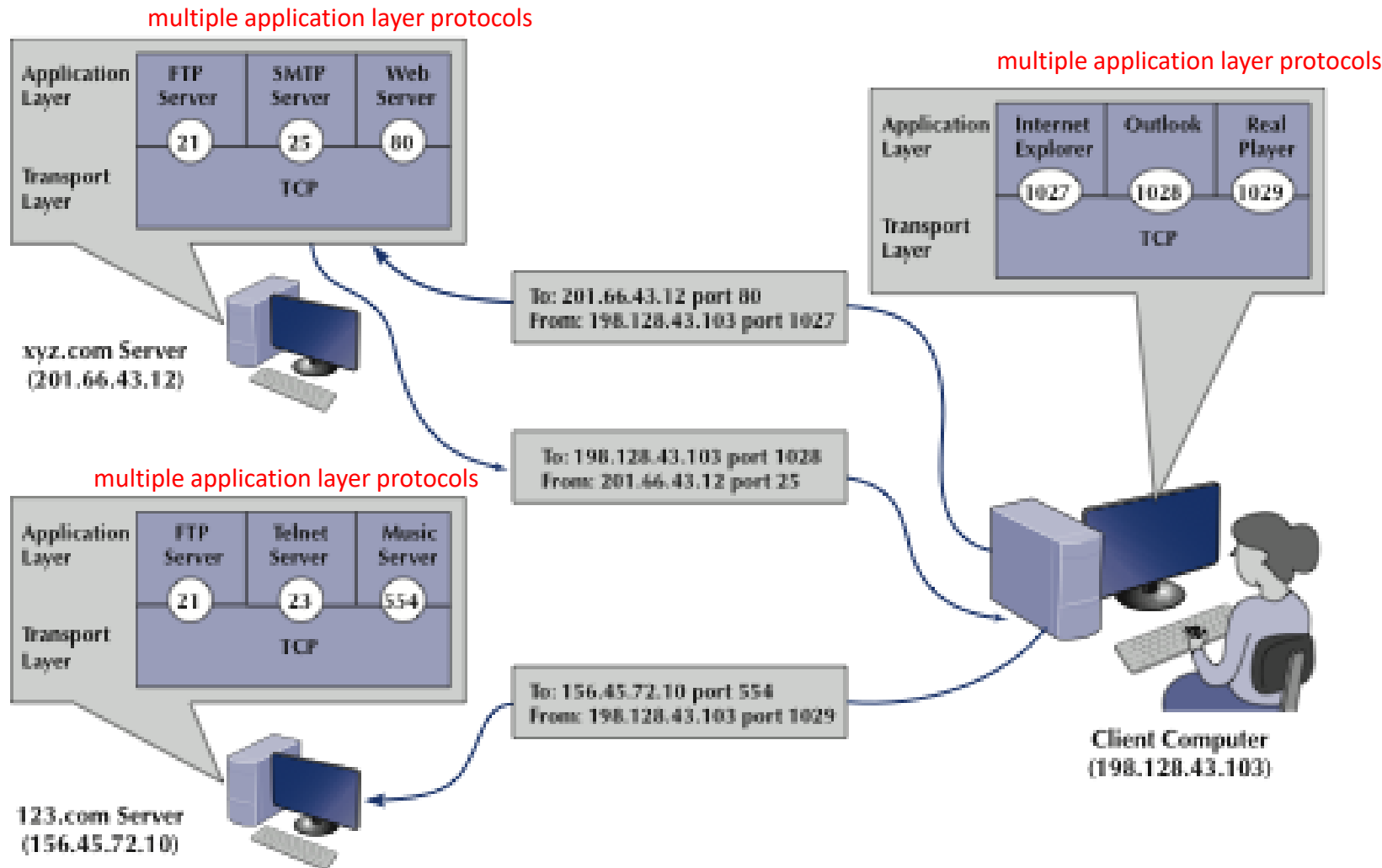


FIGURE 5-5 Linking to application layer services

Transport Layer Functions

2. Segmenting

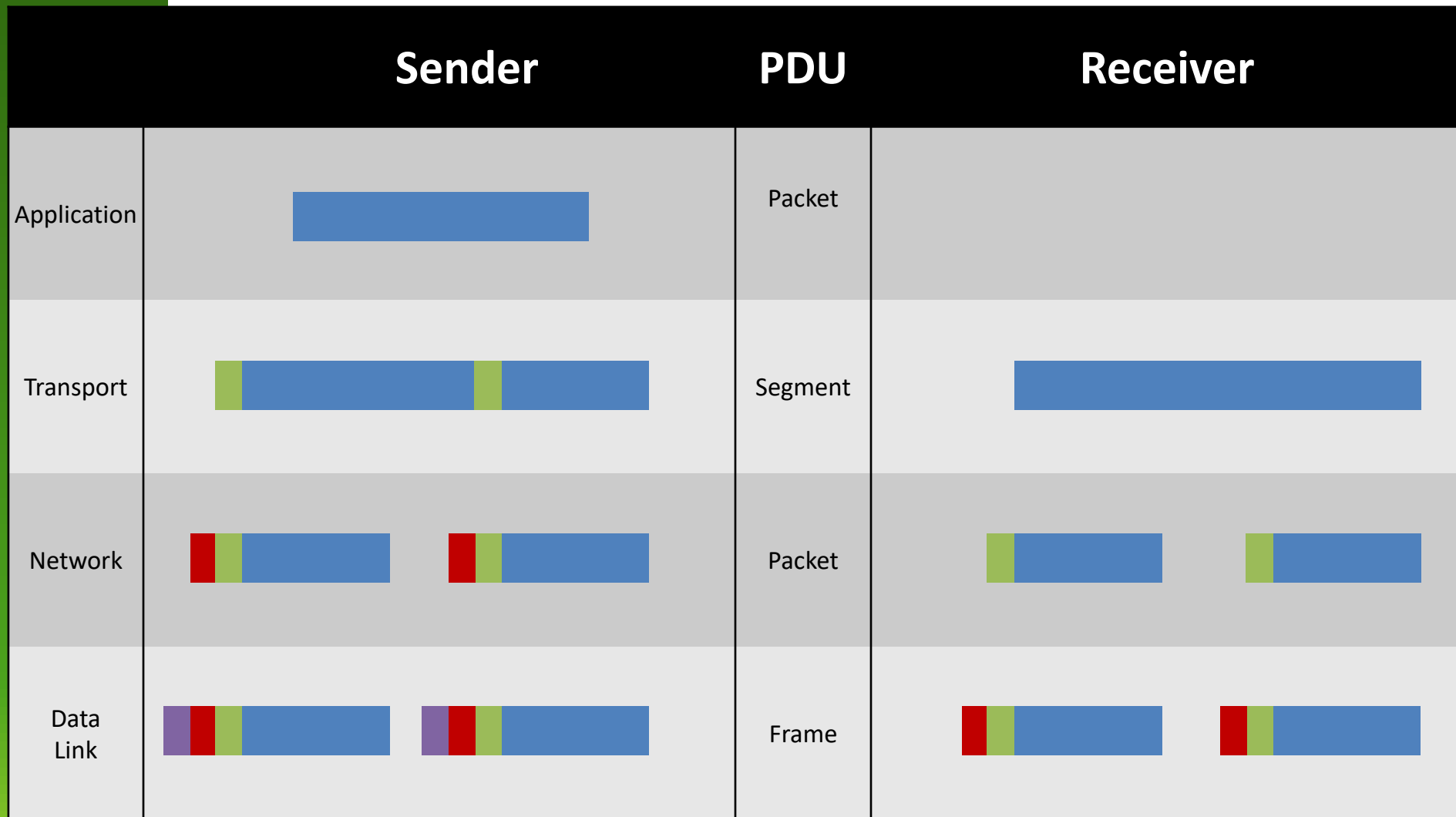
- Breaking up large files into smaller segments (and putting them back together)
- Segments may be passed individually to application layer or after reassembly
- How large are the segments?
 - Size depends on the network and **data link layer protocols**
 - Maximum Segment Size (MSS) is negotiated during TCP handshake
 - e.g., if the maximum size of the data in an Ethernet frame is 1,500 bytes and TCP and IP use 20 byte headers, the maximum segment size is 1460 bytes

Ethernet Frame Data Size → $1500 - 20 - 20 = 1460 \text{ bytes}$

↑ **TCP header**

↓ **IPv4 header**

Transport Layer Functions



Transport Layer Functions

3. Session management

- A **session** can be thought of as a **conversation between two computers** or creating a virtual circuit
- Using a session to send data is also called **connection-oriented** messaging (TCP)
- Sending messages without establishing a session is **connectionless** messaging (UDP)
- TCP connections are opened using a three-way handshake
 - SYN (synchronization characters)
 - SYN-ACK
 - ACK (acknowledgement)
- Sessions provide reliable end-to-end connections

Transport Layer Functions

3. Session management

- The transport layer software sends a special packet (SYN) to the receiver requesting that a connection be established.
- The receiver either accepts or rejects the connection, and together, they settle on the packet sizes the connection will use.
- Once the connection is established, the packets flow between the sender and the receiver, following the same route through the network.

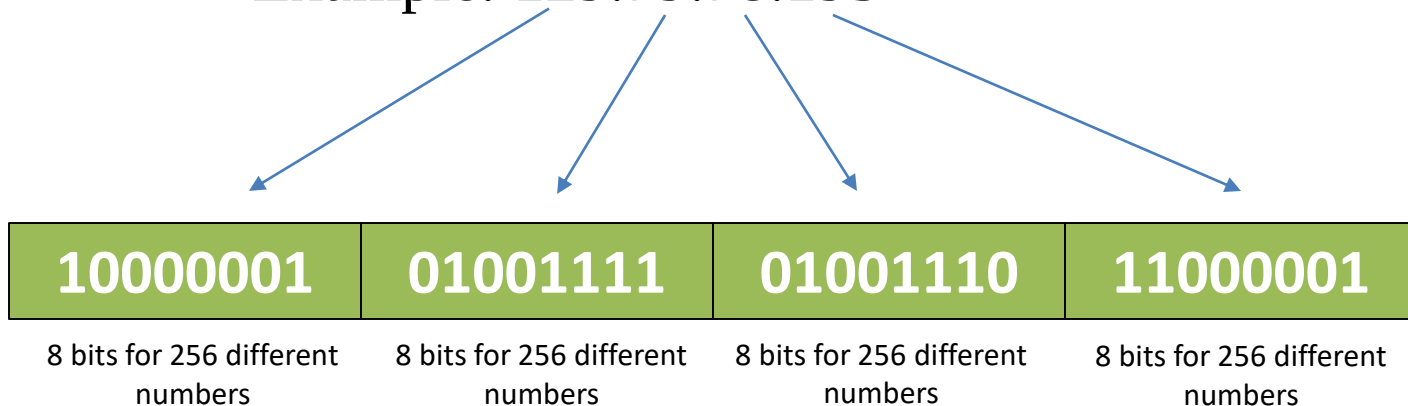
Network Layer Functions

- Addressing
 - Used to direct messages from source to destination
 - Addresses are assigned in various ways (e.g., by system administrators, ICANN, hardware vendors, etc.)
 - Addresses exist at **different layers**
 - Addresses may be **translated** from one layer to another (e.g., DNS, ARP)

Address Type	Example	Example Address
Application layer	Uniform Resource Locator (URL)	www.indiana.edu
Network layer	IP address	129.79.78.193 (4 bytes, 32 bits)
Data link layer	MAC address	1C-6F-65-F8-33-8A (6 bytes)

Network Layer Functions

- Addressing
 - IPv4 addresses are 32 bits
 - Most common way to write is using dot-decimal notation
 - Easier for people to read and remember
 - Breaks the address into four bytes and writes each byte in decimal notation instead of binary
 - Example: 129.79.78.193



Network Layer Functions

- Addressing
 - A portion of an IP address represents the **network** and the rest identifies **the host**
 - Classful addressing
 - Uses the first bits to determine number of hosts
 - Discontinued
 - Classless Inter-Domain Routing (CIDR)
 - Uses subnet masks to more flexibly divide address space into subnets
 - IP address: **129.79.78.193**
 - Subnet Mask: **255.255.255.0**

Network Layer Functions

Subnet mask

TECHNICAL FOCUS

5.1

SUBNET MASKS

Subnet masks tell computers what part of an Internet Protocol (IP) address is to be used to determine whether a destination is on the same subnet or on a different subnet. A subnet mask is a 4-byte binary number that has the same format as an IP address and is not routable on the network. A 1 in the subnet mask indicates that that position is used to indicate the subnet. A zero indicates that it is not. Therefore, a mask can only contain a continuous stream of ones.

A subnet mask of 255.255.255.0 means that the first three bytes indicate the subnet; all computers with the same first three bytes in their IP addresses are on the same subnet. This is because 255 expressed in binary is 11111111.

In contrast, a subnet mask of 255.255.0.0 indicates that the first two bytes refer to the same subnet.

Things get more complicated when we use partial-byte subnet masks. For example, suppose

the subnet mask was 255.255.255.128. In binary numbers, this is expressed as:

11111111.11111111.11111111.10000000

This means that the first three bytes plus the first bit in the fourth byte indicate the subnet address.

Likewise, a subnet mask of 255.255.254.0 would indicate the first two bytes plus the first seven bits of third byte indicate the subnet address, because in binary numbers, this is:

11111111.11111111.11111110.00000000

The bits that are ones are called *network bits* because they indicate which part of an address is the network or subnet part, whereas the bits that are zeros are called *host bits* because they indicate which part is unique to a specific computer or host.

Network Layer Functions

Subnet mask

- The subnet mask is a number that each routing device on a network can use to compare the routing packet to in order to determine if the packet stays in that portion of the network, or should be routed on to the next portion of the network.

Example:

- Network could be configured so that the **first two bytes** indicated a subnet (e.g., **128.184.x.x**).
- All computers would be given a subnet mask giving the first two bytes as the subnet indicator. (e.g., **255.255.0.0**)
- A computer with an IP address of **128.184.22.33** would be on the **same subnet** as **128.184.78.90**, and that the message stays within that network.
- If the subnet comparison shows that the message is not on that particular subnet, then **the routing device will move the packets** on in their journey.

Network Layer Functions

Every computer knows its network layer address from a configuration file that is installed when the computer is first attached to the network.

Any time a computer is moved or its network is assigned a new address, the software on each individual computer must be updated.

- **Dynamic addressing**
 - Configuring each device manually is time consuming
 - Assigning addresses permanently can be inefficient when devices are not connected to network
 - A server can **supply IP addresses automatically**
 - Dynamic Host Configuration Protocol (**DHCP**)
 - Most common protocol for dynamic addressing
 - Device sends out broadcast message
 - DHCP responds with IP settings
 - Addresses are “leased” for a length of time

Network Layer Functions

How does dynamic addressing work?

- A server is designated to supply a network layer address to a computer each time the computer connects to the network (for client computers).
- A software package is installed on the client that instructs it to contact DHCP servers using data link layer addresses.
- This message asks the servers to assign the requesting computer a unique network layer address.
- The server runs a corresponding DHCP software package that responds to these requests and sends a message back to the client giving it its network layer address (and its subnet mask).

Network Layer Functions

- Address resolution

To send a message, sender must be able to translate the application layer address (or server name) of the destination into a network layer address and in turn translate that into a data link layer address. This process is address resolution.

There are many different approaches between

- Completely decentralized (each computer is responsible for knowing all addresses)
- Completely centralized (there is one computer that knows all addresses).

Network Layer Functions

- Address resolution
 - Host (server) name resolution
 - Translate host name to IP address
 - e.g., www.indiana.edu → 129.79.78.193
 - Domain Name Service (**DNS**) servers
- DNS servers run special address databases that store thousands of Internet addresses and their corresponding IP addresses.
- These servers are in effect the "directory assistance" computers for the Internet.
- Any time a computer does not know the IP number for a computer, it sends a message to the DNS server requesting the IP number.

Network Layer Functions

How does TCP/IP perform address resolution for network layer addresses?

- TCP/IP sends a special TCP-level packet to the nearest DNS server.
- This packet asks DNS server to send the requesting computer the IP address that matches the Internet address provided.
- If DNS server has a matching name in its database, it sends back a special TCP packet with the correct IP address.
- If that DNS server does not have that Internet address in its database, it will issue the same request to another DNS server elsewhere on the Internet.
- Once your computer receives an IP address it is stored in a server address table. This way, if you ever need to access the same computer again, your computer does not need to contact a DNS server. (Most server address tables are routinely deleted whenever you turn off your computer.)

Network Layer Functions

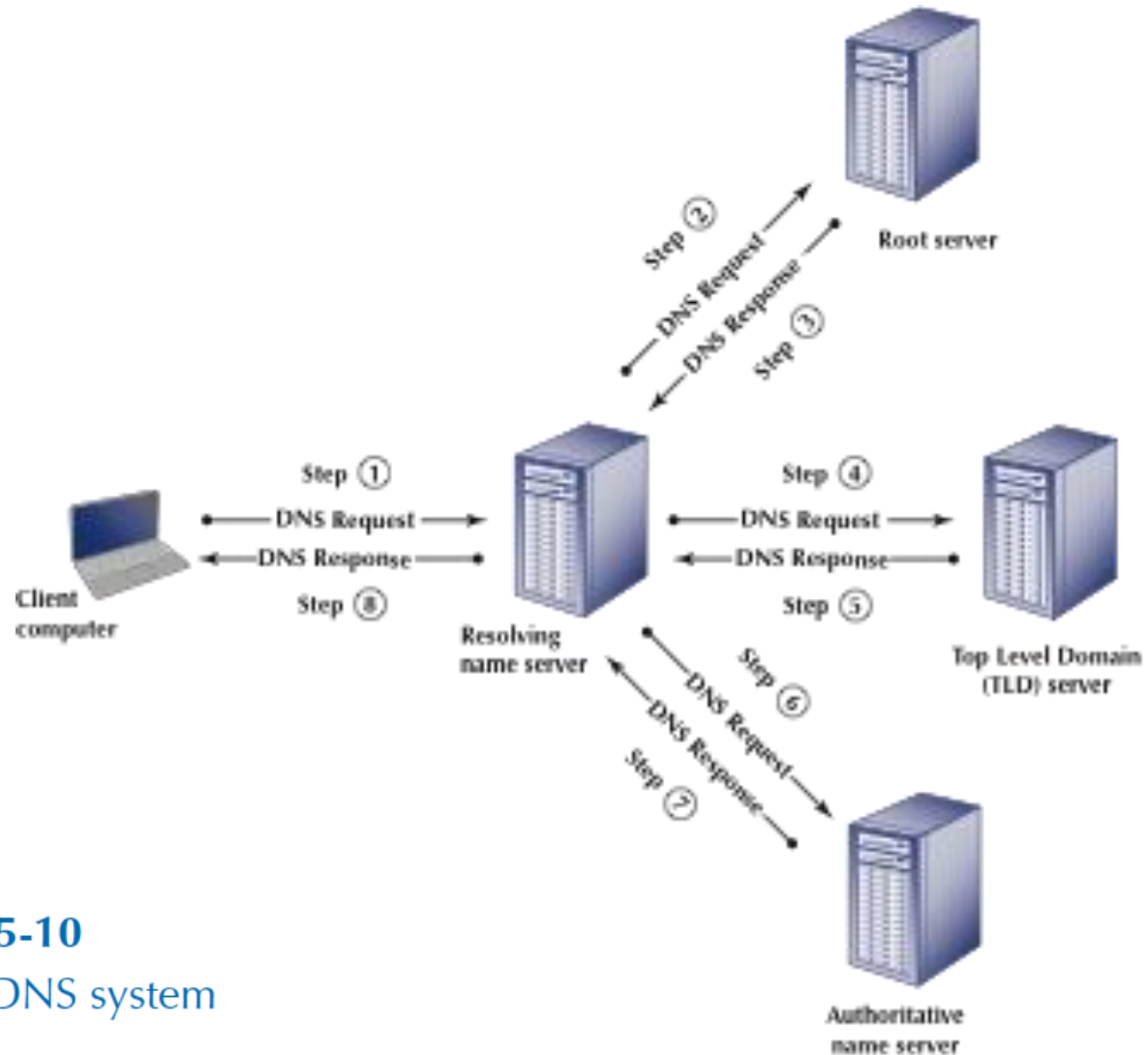


FIGURE 5-10
How the DNS system works

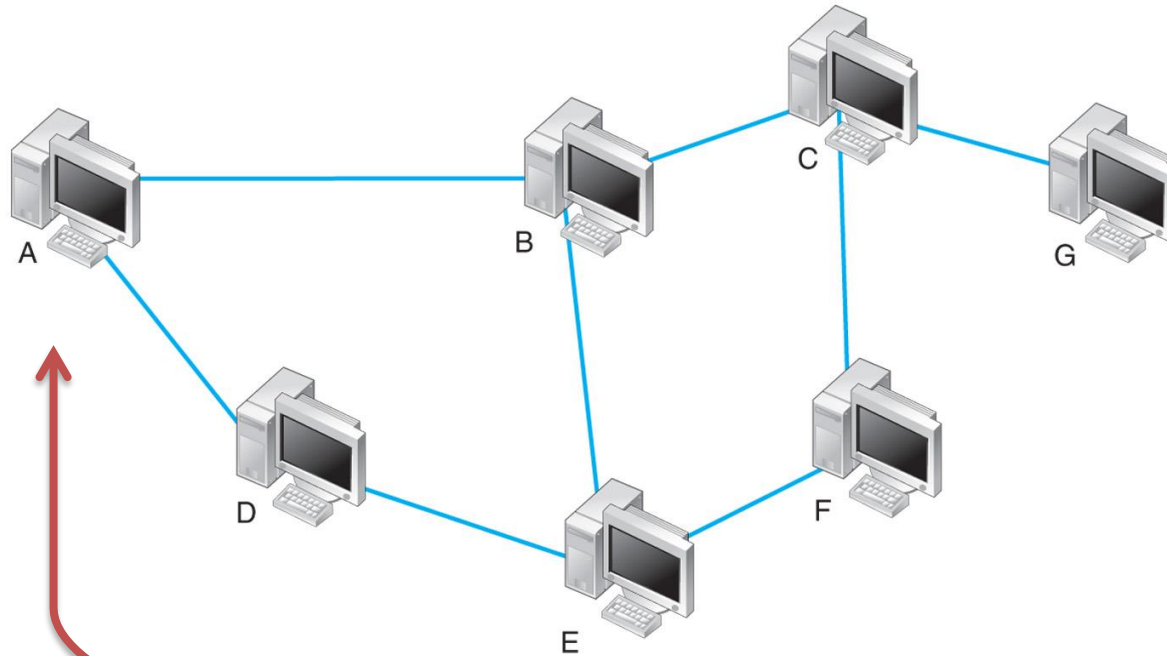
Network Layer Functions

- Routing
 - Process of identifying what path to have a packet take through a network from sender to receiver
 - Routing Tables
 - Used to make routing decisions
 - Shows which path to send packets on to reach a given destination
 - Kept by computers making routing decisions
 - Routers
 - Special purpose devices used to handle routing decisions on the Internet
 - Maintain their own routing tables

Dest.	Next
B	B
C	B
D	D
E	D
F	D
G	B

Routing

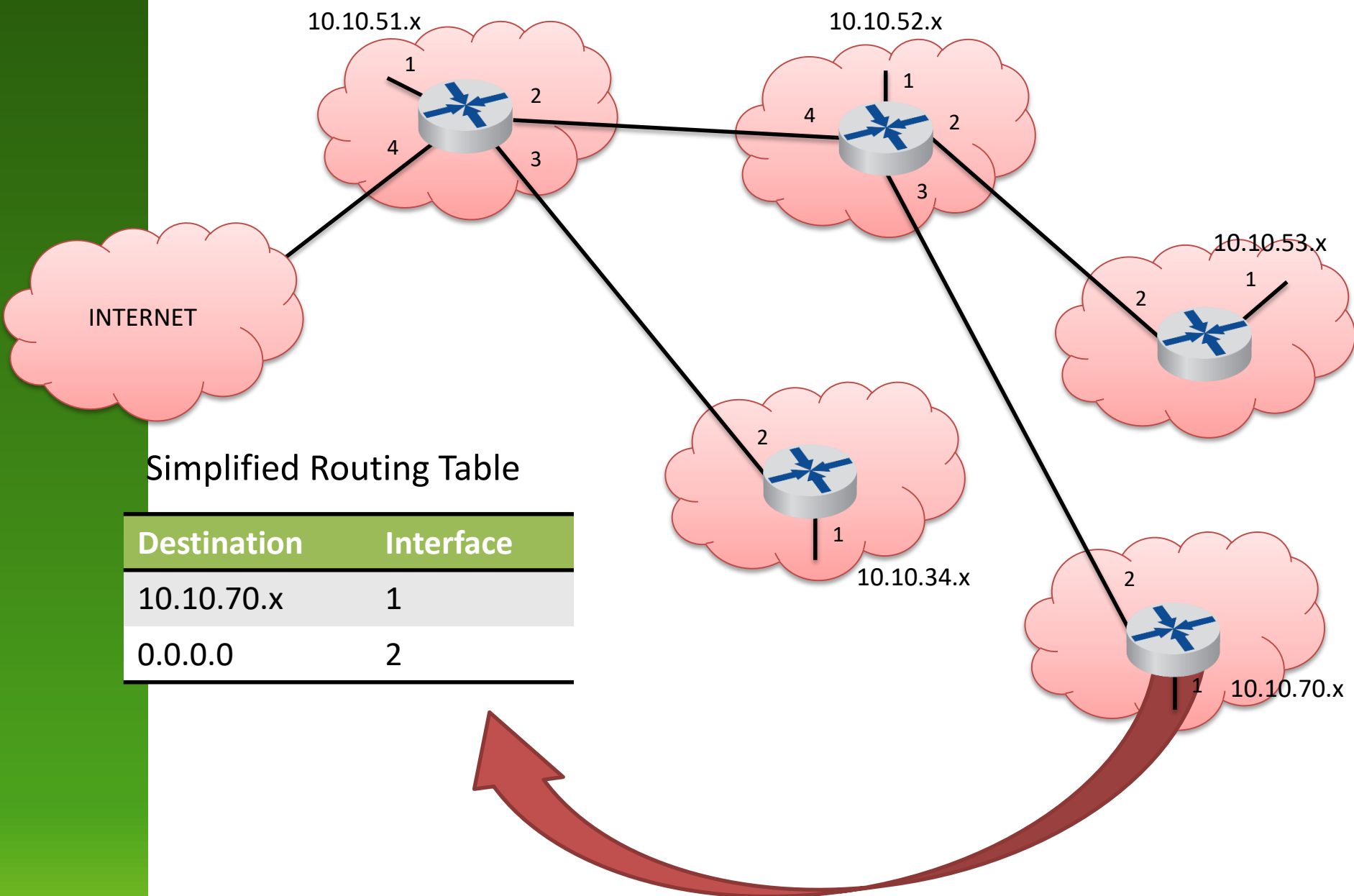
What are the possible paths from A to G?

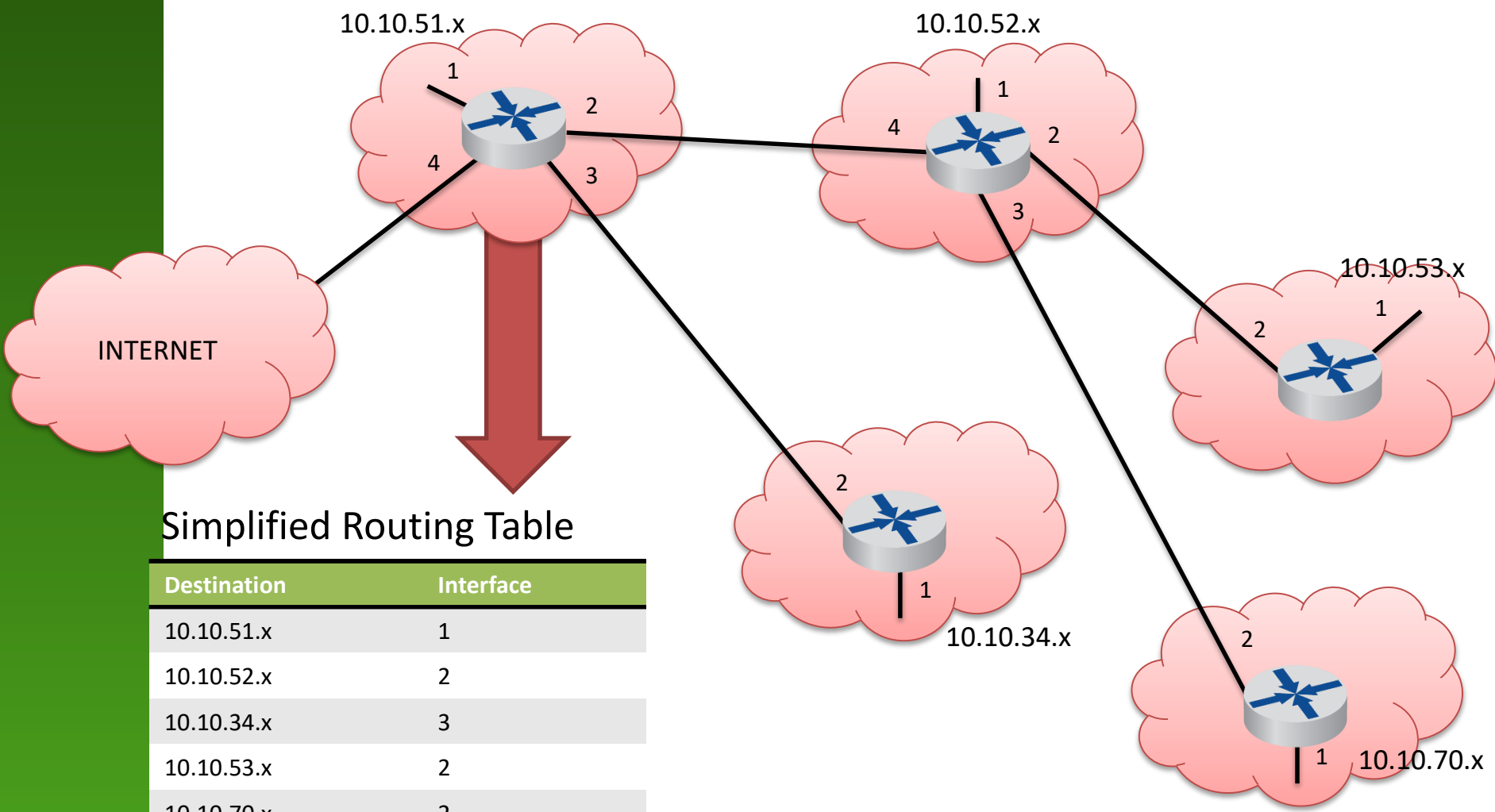


- ABCG
- ABEFCG
- ADEFCG
- ADEBCG

Simplified Routing Table for A

Dest.	Next
B	B
C	B
D	D
E	D
F	D
G	B





Destination	Interface
10.10.51.x	1
10.10.52.x	2
10.10.34.x	3
10.10.53.x	2
10.10.70.x	2
10.10.250.34	3
10.10.250.x	2
0.0.0.0	4

Routing

- **Centralized Routing**

- Routing **decisions** made by **one computer**
- Only in host-based networks (Not common anymore)
- Routing decisions are rather simple
- **All computers are connected to the central computer** by individual point-to-point circuits
- Any message received is simply retransmitted on the point-to-point circuit connected to the destination

- **Decentralized Routing**

- **Decisions** made **by each node independently** of one another
- **Information** needs to be **exchanged** to prepare routing tables
- **Automatic adjustment** when network configuration changes
- Used by the Internet

Routing

- **Static**
 - **Fixed** routing tables
 - **Manually configured** by network managers
 - **Local adjustments** when computers added or removed
- **Dynamic**
 - Routing **tables updated periodically**
 - Routers **exchange information** using **protocols** to **update tables** (may cause traffic issues)

Routing

- Dynamic routing algorithms
 - **Distance vector:** based on the number of **hops*** between two devices
 - **Link state:** based on the number of **hops, circuit speed, and traffic congestion**
(Provides more reliable, up to date paths to destinations)

Hops : The number of routers that a packet passes through from its source to its destination

Routing Protocols

A protocol used to **exchange information among computers to enable them to build and maintain their routing tables.**

When new paths are added or paths are broken or cannot be used, messages are sent among computers using a routing protocol.

Routing Information Protocol (RIP)

Open Shortest Path First (OSPF)

Enhanced Interior Gateway Routing Protocol (EIGRP)

Border Gateway Protocol (BGP)

Internet Control Message Protocol (ICMP)

Intermediate System to Intermediate System (IS-IS)

Routing Protocols

- Large networks > knowing all possible routes is impractical
- Networks are subdivided into autonomous systems of networks
- An autonomous system : network operated by one organization (one part of the Internet)
- Computers within each autonomous system know about other computers in same system and usually exchange routing info
- The routing protocols used inside an autonomous system are called **interior routing protocols**.
- Protocols used between autonomous systems are **exterior routing protocols**.

Routing Protocols

- **Routing Information Protocol (RIP)**
 - Dynamic **distance vector** protocol used for **interior** routing
 - Operation
 - Network **manager builds** the routing table
 - Routing **tables broadcast periodically** (e.g., every min.)
 - When new computers are added, **router counts “hops”** and selects the **shortest route**
 - **Useful in smaller, less complex** networks

Routing Protocols

- **Open Shortest Path First (OSPF)**
 - Dynamic **link state** protocol used for **interior** routing
 - **Most widely used** interior routing protocol on **large enterprise networks**
 - More reliable paths
 - **Less burdensome** to the network because **only updates sent**
(no periodic update as RIP)

Routing Protocols

- **Enhanced Interior Gateway Routing Protocol (EIGRP)**
 - A dynamic **link state** protocol (developed by Cisco)
 - Records **transmission capacity, delay time, reliability** and load for all paths
 - Keeps the **routing tables for its neighbors** and **uses this information in its routing decisions** as well

Routing Protocols

- **Intermediate System to Intermediate System (IS-IS)**
 - Dynamic **link state** protocol used for **interior** routing
 - Commonly used in **large networks**
 - IS-IS is an ISO protocol that has been added to many TCP/IP networks.

Routing Protocols

- **Internet Control Message Protocol (ICMP)**
 - **Simplest interior** routing protocol on the Internet
 - An error-reporting protocol that enables computers to **report routing errors** to message senders
 - Has very **limited ability to update routing tables**

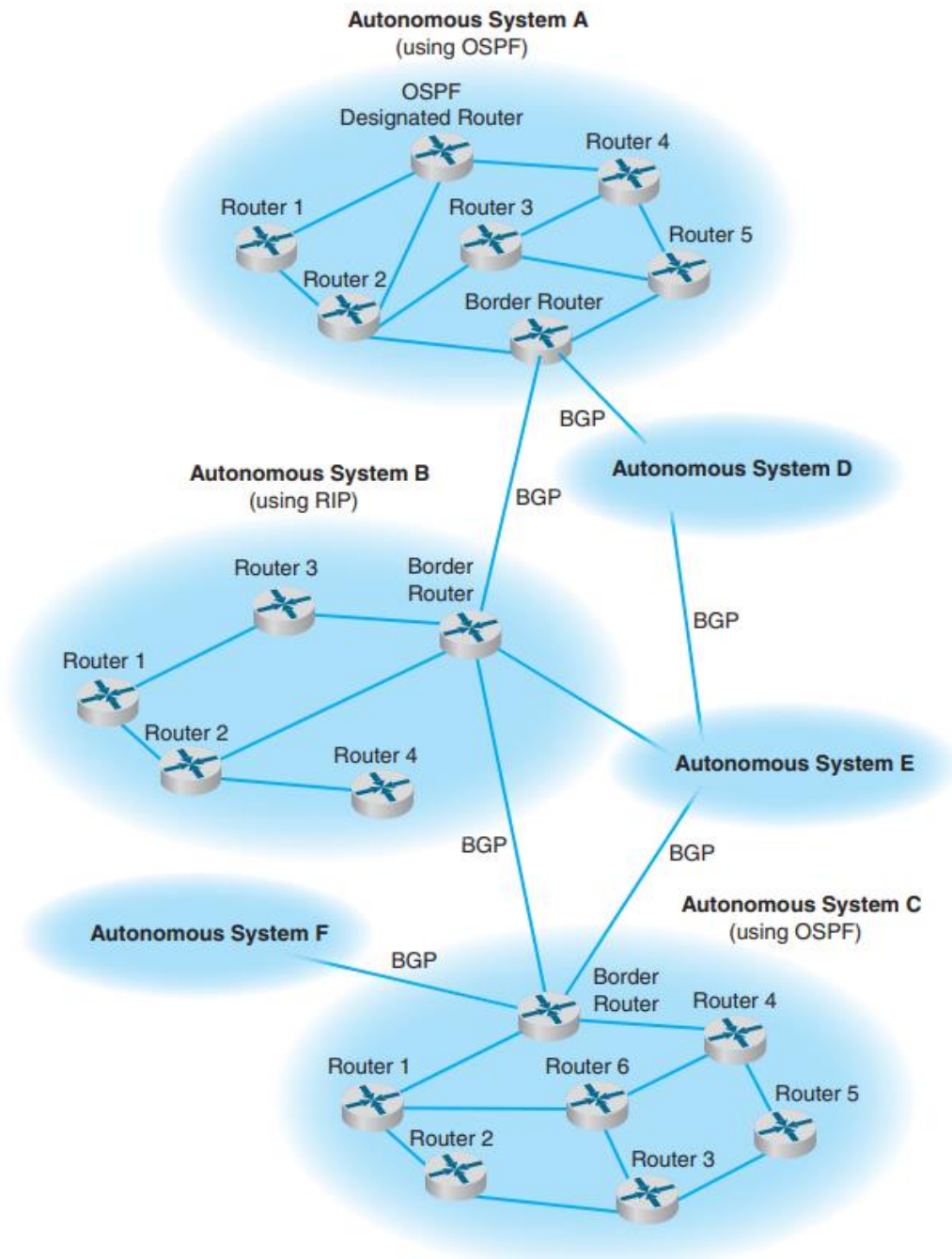
Routing Protocols

If each network uses a **different protocol** internally, how are they able to communicate?

- **Border Gateway Protocol (BGP)**
 - Dynamic **distance vector** protocol used for **exterior** routing
 - Far **more complex than interior** routing protocols
 - Provide routing info only on selected routes (e.g., preferred or best route)

FIGURE 5-13

Routing on the Internet
with Border Gateway
Protocol (BGP), Open
Shortest Path First
(OSPF), and Routing
Information Protocol
(RIP)



Types of messages in a network

- **Unicast**

One computer sends a message to another computer (e.g., a client requesting a Web page)

- **Broadcast**

The message is sent to all computers on a specific LAN or subnet (one computer to all computers in the network)

- **Multicast**

Types of messages in a network

- **Multicast**

One computer sends a message to a group of computers (e.g., video conference)

Same data needs to reach multiple receivers and avoid transmitting it once for each receiver

- Useful when access link has bandwidth limitations
- Many implementations at different layers
- Hosts dynamically join and leave multicast groups using Internet Group Management Protocol (IGMP)

Types of messages in a network

- **How does Multicasting work?**

Computers wishing to participate in a multicast (e.g., for videoconferencing) **send a message to the host computer** or other routing computer along the way using a special type of TCP-level packet called Internet Group Management Protocol (IGMP).

Each multicast group is **assigned a special Class D IP** address to identify the group.

Any computer performing routing knows to **route all multicast messages with this Class D IP** address onto the subnet that contains the requesting computer.

The **routing computer sets the data link layer address on multicast messages to a matching multicast data link layer address.**

Each **requesting computer must inform its data link layer software to process incoming messages with this multicast data link layer address.**

When the **multicast session ends, client computer sends another IGMP message to the organizing computer** or the computer performing routing to **remove it from the multicast group.**

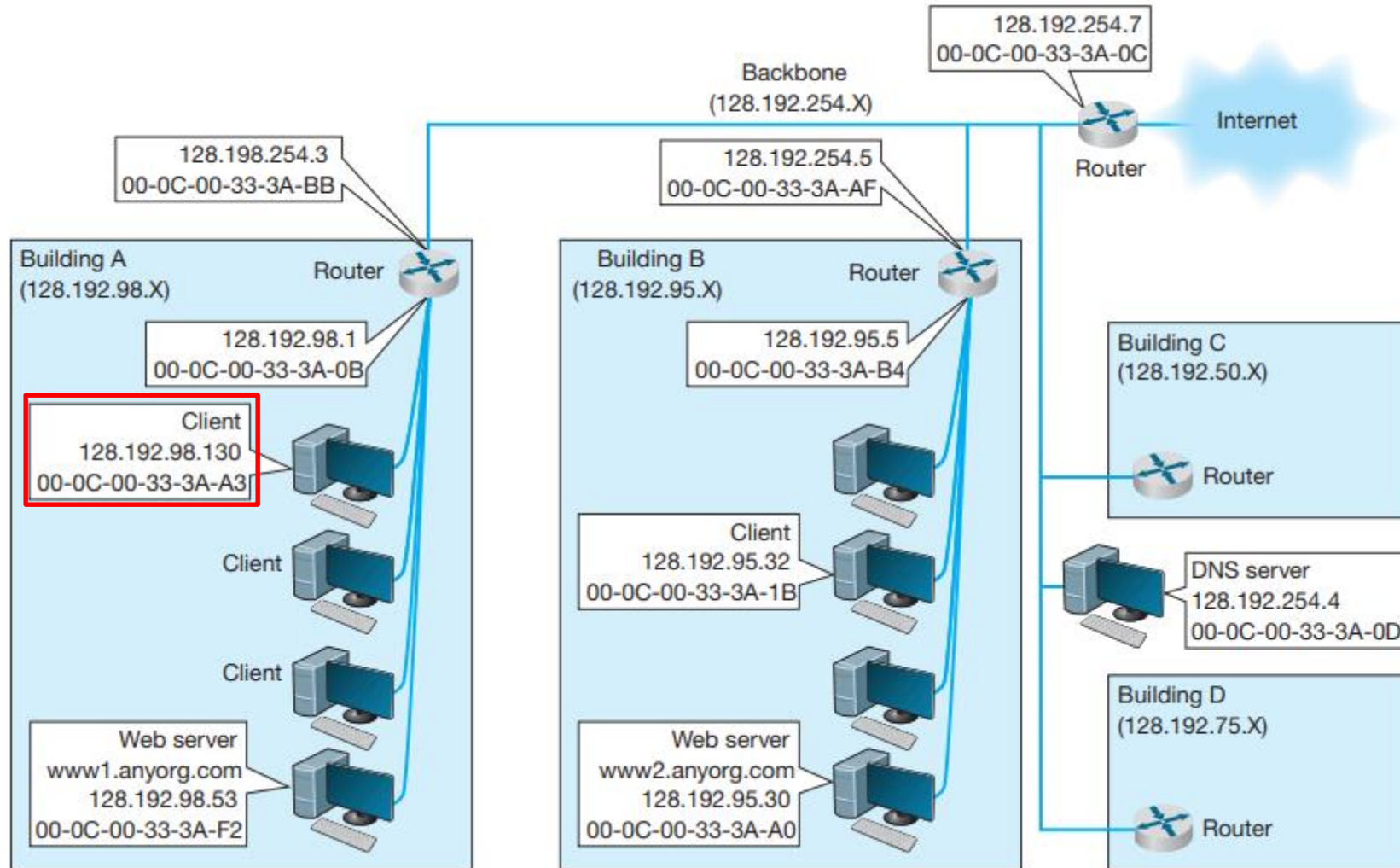
TCP/IP and Layers

- Host Computers
 - Packets move through all layers
- Gateways, Routers
 - Packet moves from Physical layer to Data Link Layer through the network Layer
- At each stop along the way
 - Ethernet packets is removed and a new one is created for the next node
 - IP and above packets never change in transit (created by the original sender and destroyed by the final receiver)

TCP/IP Example (pp. 174-179)

- Required network addressing information:
 1. Device's own IP address
 2. Subnet mask
 3. IP address of default gateway (most commonly router)
 4. IP address of at least one DNS server
- Obtained from a configuration file or DHCP

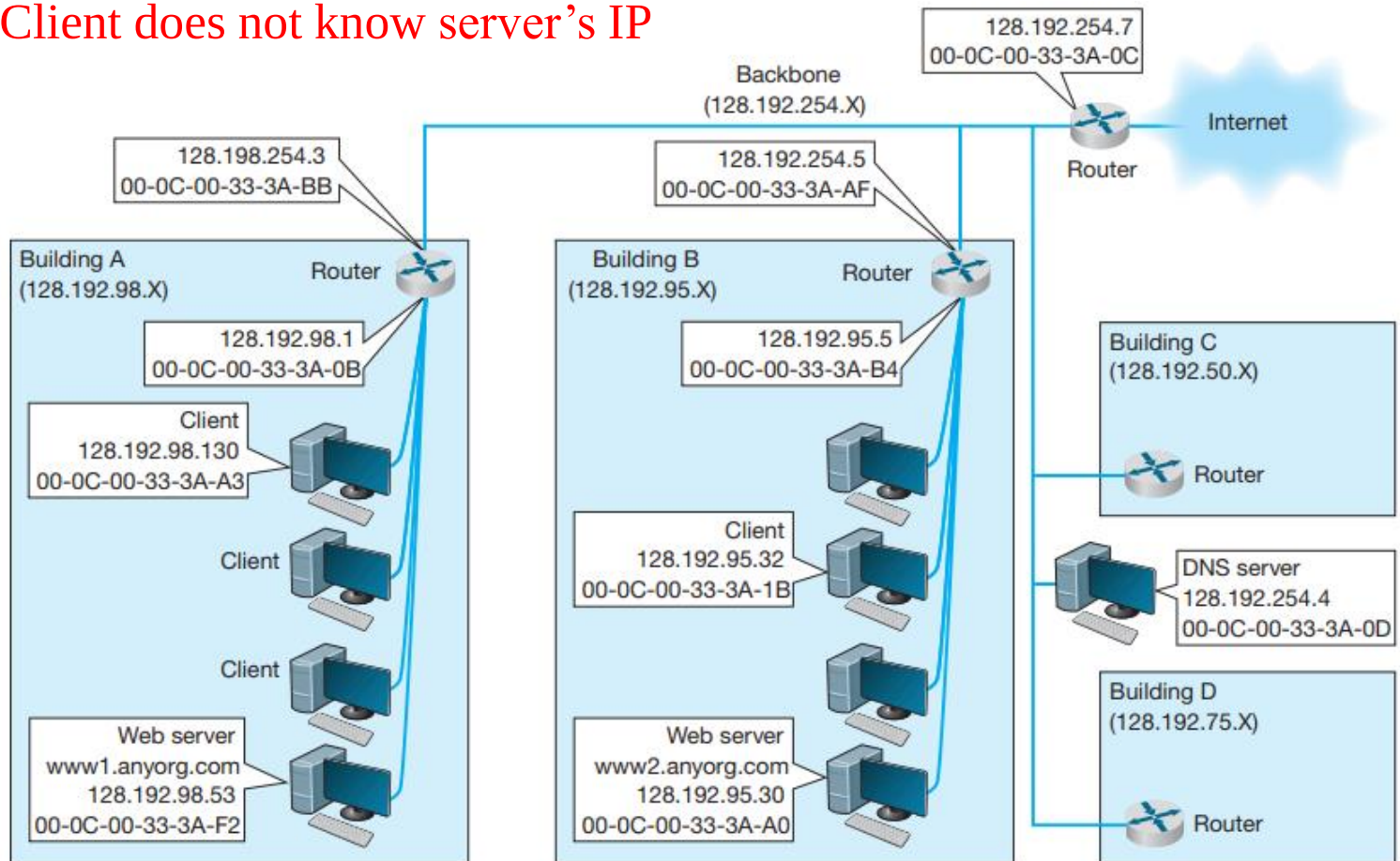
TCP/IP Example



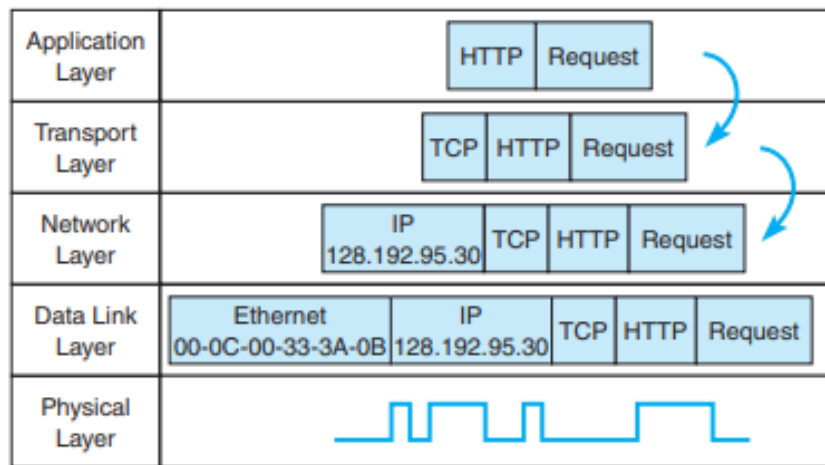
- Suppose we have an HTTP request from Client in building A.

TCP/IP Examples

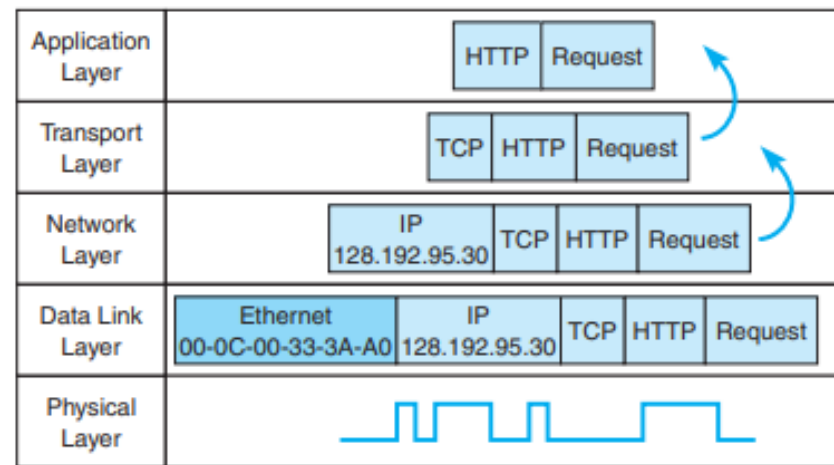
1. A Client (128.192.98.130) requests a Web page from a server (www1.anyorg.com). **Client knows the server's IP**
2. Same client requests a Web page from a server (www2.anyorg.com) on a different subnet. **Client knows the server's IP**
3. Same client requests a Web page from a server (www1.anyorg.com). **Client does not know server's IP**



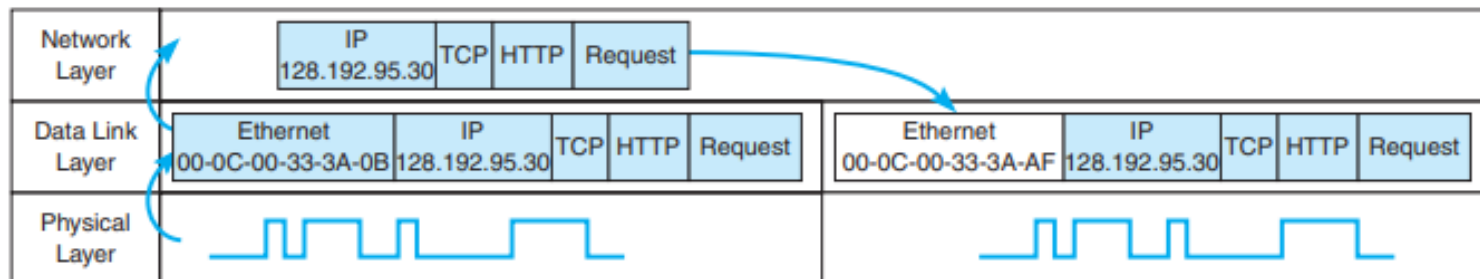
Sender (Client in Building A)



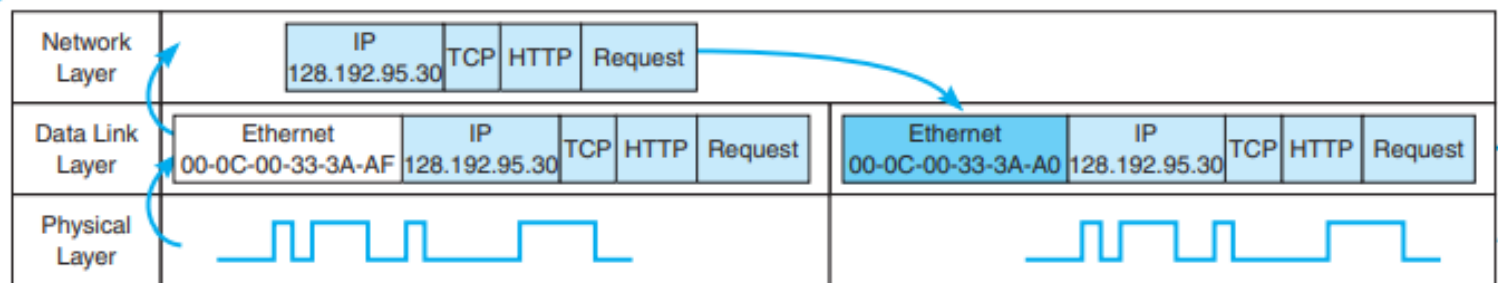
Receiver (Server in Building B)



Gateway (Router in Building A)



Gateway (Router in Building B)

**FIGURE 5-18** How messages move through the network layers.

Note: The addresses in this example are destination addresses

TCP/IP Examples (Known Addresses, Same Subnet)

1. A Client (128.192.98.130) requests a Web page from a server (www1.anyorg.com). **Client knows the server's IP**

The application layer software (i.e., Web browser) passes an HTTP packet containing the user request to the transport layer software requesting a page from 128.192.98.53. The transport layer software (TCP) would take the HTTP packet, add a TCP segment, and then hand it to the network layer software (IP). The network layer software will compare the destination address (128.192.98.53) to the subnet mask (255.255.255.0) and discover that this computer is on its own subnet. The network layer software will then search its data link layer address table and find the matching data link layer address (00-0C-00-33-3A-F2). The network layer would then attach an IP packet and pass it to the data link layer, along with the destination Ethernet address. The data link layer would surround the frame with an Ethernet frame and transmit it over the physical layer to the Web server (Figure 5.15). The data link layer on the Web server would perform error checking before passing the HTTP packet with the TCP segment and IP packet attached to its network layer software. The network layer software (IP) would then process the IP packet, see that it was destined to this computer, and pass it to the transport layer software (TCP).

TCP/IP Examples (Known Addresses, Same Subnet)

1. A Client (128.192.98.130) requests a Web page from a server (www1.anyorg.com). **Client knows the server's IP**

This software would process the TCP segment, see that there was only one packet, and pass the HTTP packet to the Web server software. The Web server software would find the page requested, attach an HTTP packet, and pass it to its transport layer software. The transport layer software (TCP) would break the Web page into several smaller segment, each less than 1,500 bytes in length, and attach a TCP segment (with a number to indicate the order) to each. Each smaller segment would then go to the network layer software, get an IP packet attached that specified the IP address of the requesting client (128.192.98.130), and be given to the data link layer with the client's Ethernet address (00-0C-00-33-3A-A3) for transmission. The data link layer on the server would transmit the frames in the order in which the network layer passed them to it. The client's data link layer software would receive the frames, perform error checking, and pass the IP packets inside them to the network layer. The network layer software (IP) would check to see that the packets were destined for this computer and pass the TCP segments they contained to the transport layer software. The transport layer software (TCP) would assemble the separate segments, in order, back into one Web page, and pass the HTTP packet in turn to the Web browser to display on the screen.

TCP/IP Examples (Known Addresses, Different Subnet)

2. Same client requests a Web page from a server (www2.anyorg.com) on a different subnet. **Client knows the server's IP**

Application layer software would pass an HTTP packet to the transport layer software (TCP) with the Internet address of the destination www2.anyorg.com: 128.192.95.30. The transport layer software (TCP) would make sure that the request fit in one segment and hand it to the network layer. The network layer software (IP) would then check the subnet mask and would recognize that the Web server is located outside of its subnet. Any messages going outside the subnet must be sent to the router (128.192.98.1), whose job it is to process the message and send the message on its way into the outside network.

The network layer software would check its address table and find the Ethernet address for the router. It would therefore set the data link layer address to the router's Ethernet address on this subnet (00-0C-00-33-3A-0B) and pass the IP packet to the data link layer for transmission. The data link layer would add the Ethernet frame and pass it to the physical layer for transmission.

TCP/IP Examples (Known Addresses, Different Subnet)

2. Same client requests a Web page from a server (www2.anyorg.com) on a different subnet. **Client knows the server's IP**

The router would receive the message and its data link layer would perform error checking and send an acknowledgement before passing the packet to the network layer software (IP). The network layer software would read the IP address to determine the final destination. The router would recognize that this address (128.192.95.30) needed to be sent to the 128.192.95.x subnet. It knows the router for this subnet is 128.192.254.5. It would pass the packet back to its data link layer, giving the Ethernet address of the router (00-0C-00-33-3A-AF).

This router would receive the message (do error checking, etc.) and read the IP address to determine the final destination. The router would recognize that this address (128.192.95.30) was inside its 128.192.95.x subnet and would search its data link layer address table for this computer. It would then pass the packet to the data link layer along with the Ethernet address (00-0C-00-33-3A-A0) for transmission.

TCP/IP Examples (Known Addresses, Different Subnet)

2. Same client requests a Web page from a server (www2.anyorg.com) on a different subnet. **Client knows the server's IP**

The www2.anyorg.com web server would receive the message and process it. This would result in a series of TCP/IP packets addressed to the requesting client (128.192.98.130). These would make their way through the network in reverse order. The Web server would recognize that this IP address is outside its subnet and would send the message to the 128.192.95.5 router using its Ethernet address (00-0C-00-33-3A-B4). This router would then send the message to the router for the 128.192.98.x subnet (128.192.254.3) using its Ethernet address (00-0C-00-33-3A-BB). This router would in turn send the message back to the client (128.192.98.130) using its Ethernet address (00-0C-00-33-3A-A3).

This process would work in the same way for Web servers located outside the organization on the Internet. In this case, the message would go from the client to the 128.192.98.x router, which would send it to the Internet router (128.192.254.7), which would send it to its Internet connection. The message would be routed through the Internet, from router to router until it reached its destination. Then the process would work in reverse to return the requested page.

TCP/IP Examples (Unknown Addresses)

3. Same client requests a Web page from a server (www1.anyorg.com) but **Client does not know the server's IP**

The Web browser realizes that it does not know the IP address after searching its IP address table and not finding a matching entry. Therefore, it issues a DNS request to the name server (128.192.254.4). The DNS request is passed to the transport layer (TCP), which attaches a UDP datagram and hands the message to the network layer.

Using its subnet mask, the network layer (IP) will recognize that the DNS server is outside of its subnet. It will attach an IP packet and set the data link layer address to its router's address.

The router will process the message and recognize that the 128.192.254.4 IP address is on the BN. It will transmit the packet using the DNS server's Ethernet address.

The name server will process the DNS request and send the matching IP address back to the client via the 128.198.98.x subnet router.

TCP/IP Examples (Unknown Addresses)

3. Same client requests a Web page from a server (www1.anyorg.com) but **Client does not know the server's IP**

The IP address for the desired computer makes its way back to the application layer software, which stores it in its IP table. It then issues the HTTP request using the IP address for the Web server (128.192.98.53) and passes it to the transport layer, which in turn passes it to the network layer. The network layer uses its subnet mask and recognizes that this computer is on its subnet. However, it does not know the Web server's Ethernet address. Therefore, it broadcasts an ARP request to all computers on its subnet, requesting that the computer whose IP address is 128.192.98.53 to respond with its Ethernet address.

This request is processed by all computers on the subnet, but only the Web server responds with an ARP packet giving its Ethernet address. The network layer software on the client stores this address in its data link layer address table and sends the original Web request to the Web server using its Ethernet address.

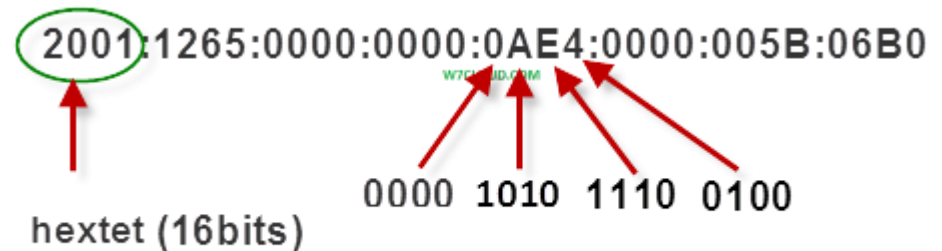
This process works the same for a Web server outside the subnet, whether in the same organization or anywhere on the Internet. If the Web server is far away (e.g., Australia), the process will likely involve searching more than one name server, but it is still the same process

Implications for Management

- Organizations standardizing on TCP/IP
 - Decreases costs of equipment and training
 - Network providers are also moving towards standardization
- Slow transition to IPv6

Example of IPv6

Hexadecimal #	Binary Representation
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001
A	1010
B	1011
C	1100
D	1101
E	1110
F	1111



IPv6 Address Example

$$8 \times 16 = 128 \text{ bits}$$